



Data Science & Artificial Intelligence

Machine Learning

Nearest Neighbour





Topics to be Covered



- Topic ... (Nearest Neighbour algo)
- Topic ...
- Topic ...
- Topic ...

* Work book

→ Telegnam Siddhath Sin pu

→ 25 July msg

Solved wrs...

→ Champion $\Rightarrow$ linear Regression

Concept

Overfit

Rattu tota

- Book given



- If Q is from book

100% accurate

- New Question

low accuracy

Underfit

Kuch nahi padhta



Low accuracy

low accuracy

Best fit

Samajh Samajh ke padhta hai



high accuracy but not 100%

Best accuracy

Bias $\longleftrightarrow$ proxy to training error... Overfit Bias 0

Underfit high

Best Fit normal

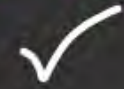
Variance

Concept

Overfit

Rattu tota

- Book given



- If Q is from book

100% accurate

- New Question

low accuracy

Underfit

Kuch nahi padhta

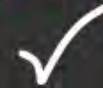


Low accuracy

low accuracy

Best fit

Samajh Samajh ke padhta hai



high accuracy but not 100%

Best accuracy

• Variance $\Rightarrow$ The change in result when test point is slightly shifted or varied.

Overfit: γ large

Underfit
Zero

Best Fit
nominal value... ✓

• Variance $\Rightarrow$ The change in result when test point is slightly shifted or varied.

In which model?

If test point is slightly shifted...

then y changes badly...

Regression

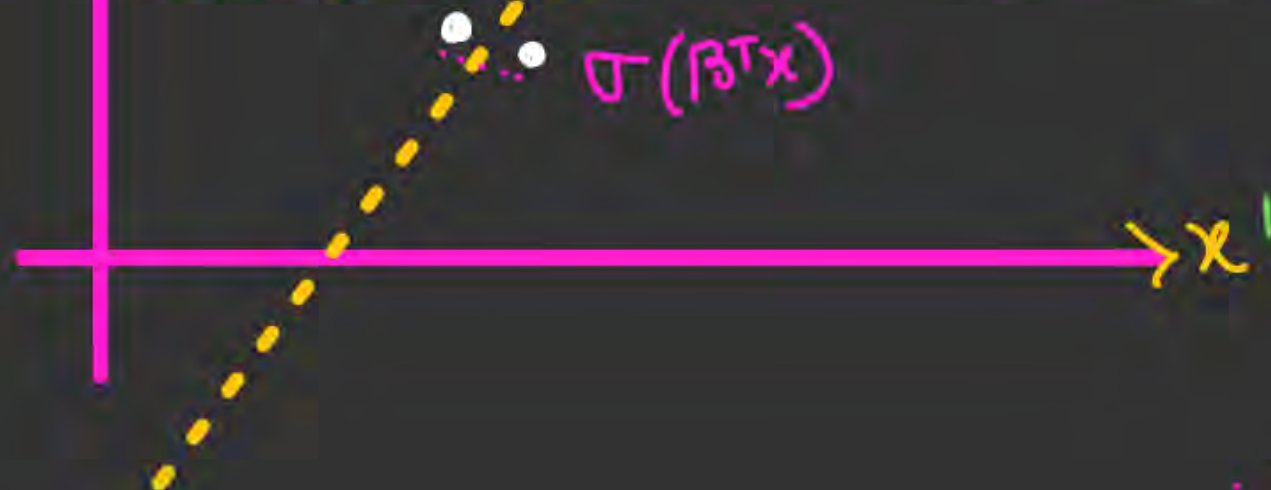
x^2

$$h = \sigma(\beta^T x)$$

$$\sigma(\beta^T x)$$

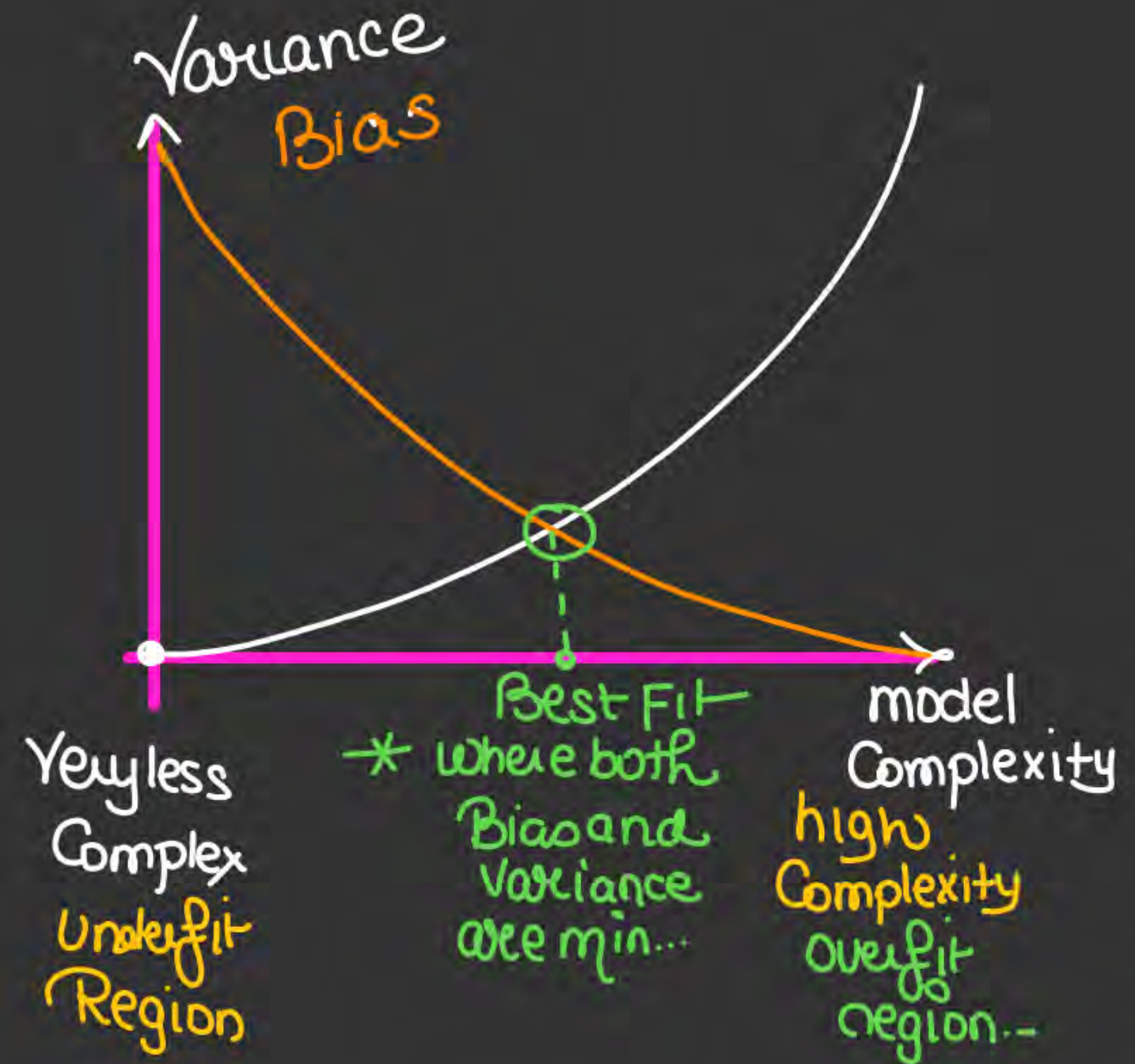
linear classifier

$$10^7 x^1 + 10^8 x^2 + \beta_0 = 0$$



Important

	Over Fit	Under Fit	Best Fit
1) Bias	Zero	high	normal
2) Training error	Zero	high	normal
3) Variance	V. high	Zero	normal
4) Testing error	higher.	V. high	least



[MCQ]



#Q. Increasing the regularizing coefficient value for a ridge regressor will

- (i) Increase or maintain model bias.
- (ii) Decrease model bias.
- (iii) Increase or maintain model variance.
- (iv) Decrease model variance.

$\lambda \uparrow$ [overfit $\rightarrow$ underfit]

$$RR: \min(RSS + \lambda \sum \beta_i^2)$$

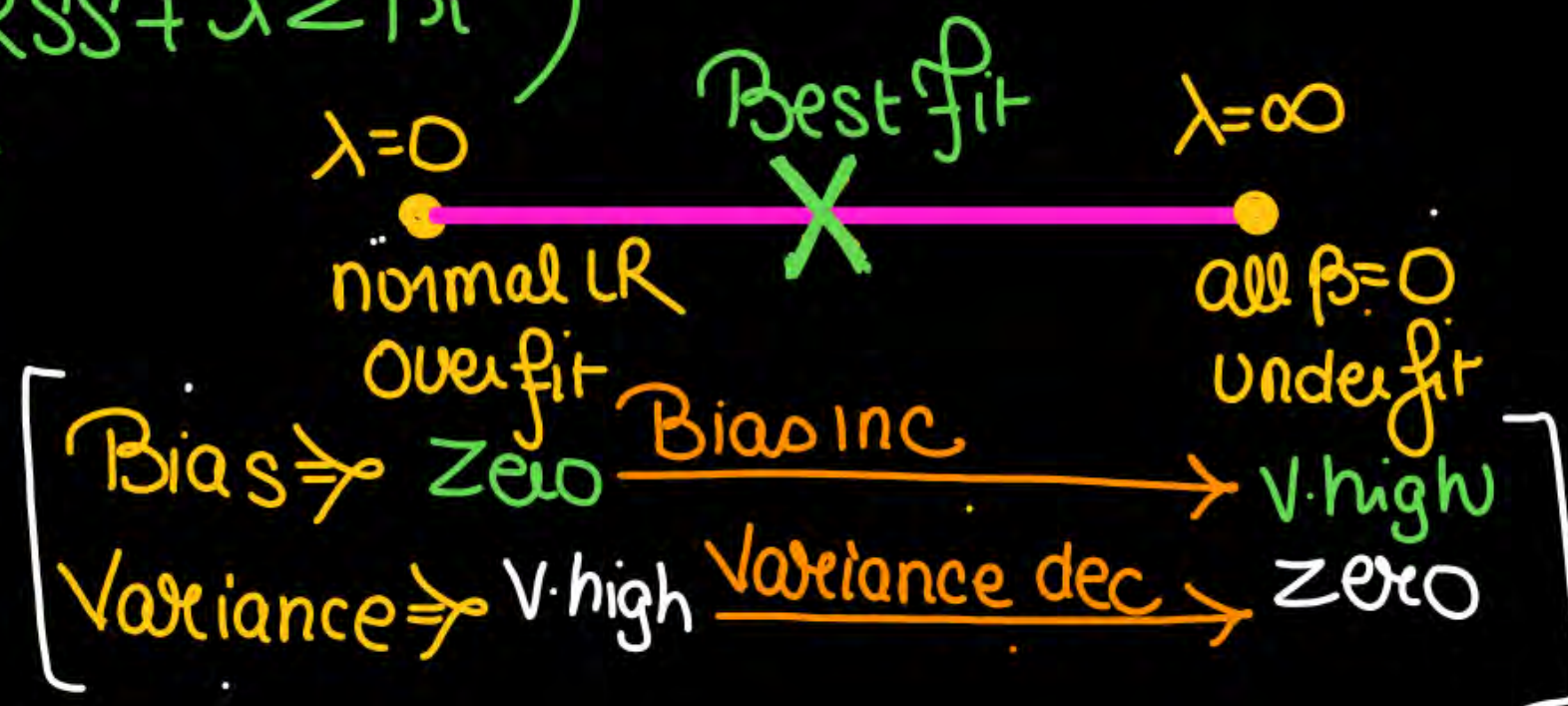
$\lambda \uparrow$

A i and iii

B i and iv

C ii and iii

D ii and iv



[MCQ]



#Q. Which among the following may help to reduce overfitting demonstrated by a model

✓ i. Change the loss function. ^{Yes ✓}

✓ ii. Reduce model complexity.

✓ iii. Increase the training data.

iv. Increase the number of optimization routine steps.

(model try to capture the pattern of the data)

A ii and i

B ii and iii

C i, ii, and iii

D i, ii, iii, and iv

• model is overfitting $\Rightarrow$
1) loss fcn change?? ^{yes ✓}
LR $\rightarrow$ RR
2) Complexity $\downarrow$, overfitting reduce ✓
3) Training data inc,

[MCQ]



#Q. Which among the following may help to reduce overfitting demonstrated by a model

i. Change the loss function. ^{Yes ✓}

ii. Reduce model complexity.

iii. Increase the training data.

iv. Increase the number of optimization routine steps.

Yes ✓

(model try to capture the pattern of the data)

→ we use gradient descent to minimize any loss fcn...

→ if we inc No of iteration $\Rightarrow$ loss fcn ko zero karke hi manunga...

→ model will give zero error on training data...

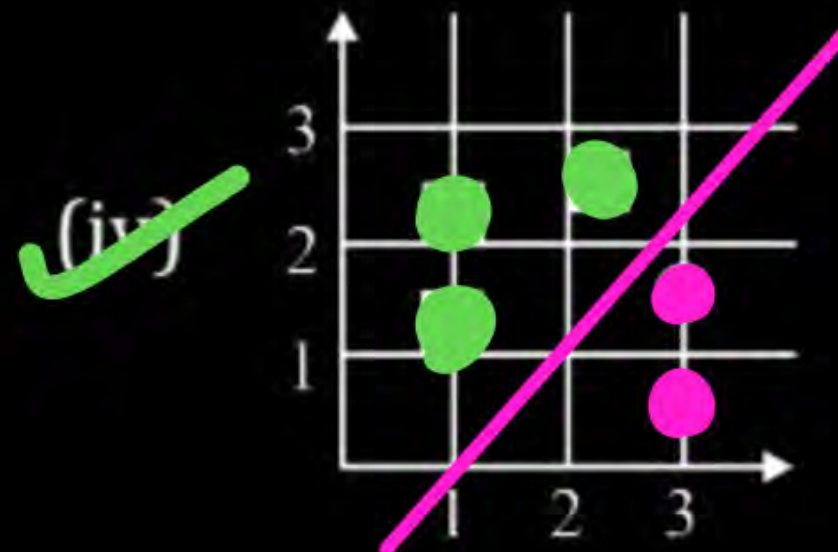
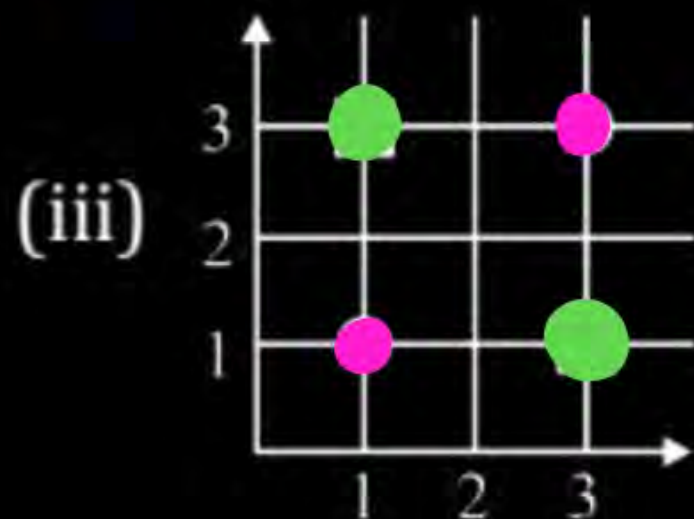
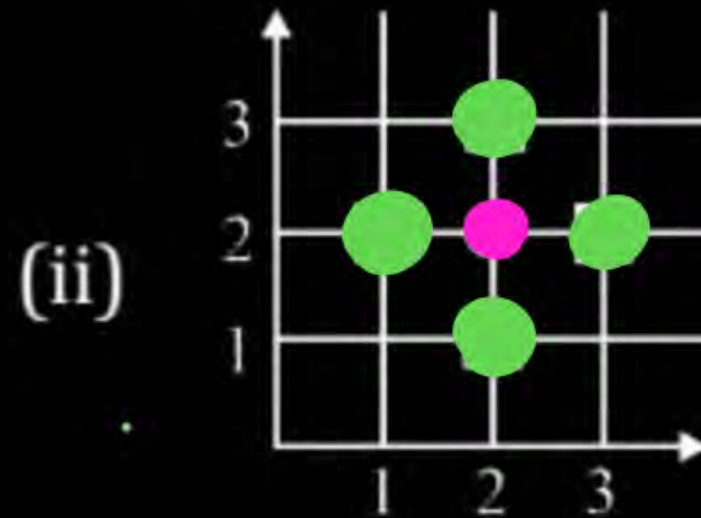
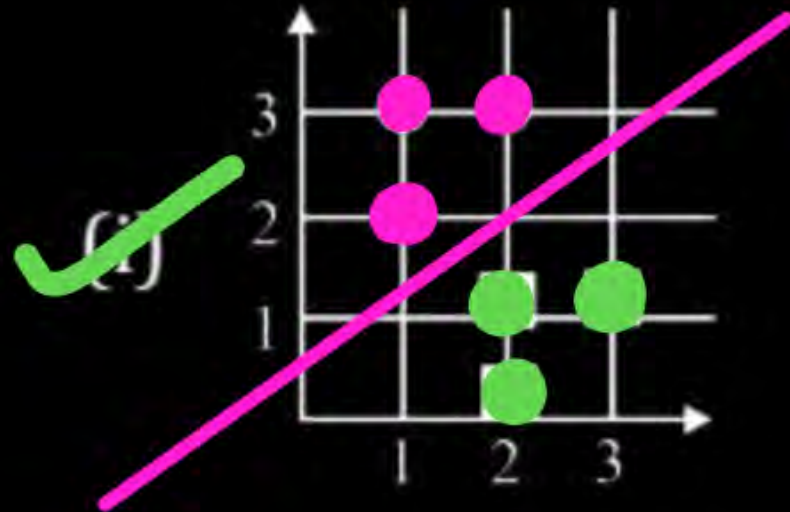
→ inc overfitting..

[MCQ]



#Q. Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and square

Which of the following is/are TRUE?



Continue...

Ch: 3

Nearest Neighbour algorithm

NN: algorithm...

- we believe that the neighbours are similar to each other...

- How we find neighbour of a point



- Find closest neighbour of Point A:—

* Find distance of point A from all other points

* Jiska min distance hoga wahi hoga closest neighbour

3D....

$$x_1 = (x_1^1, x_1^2, x_1^3)$$

$$x_2 = (x_2^1, x_2^2, x_2^3)$$

Euclidean distance...

$$d_{12} = \sqrt{(x_1^1 - x_2^1)^2 + (x_1^2 - x_2^2)^2 + (x_1^3 - x_2^3)^2}$$

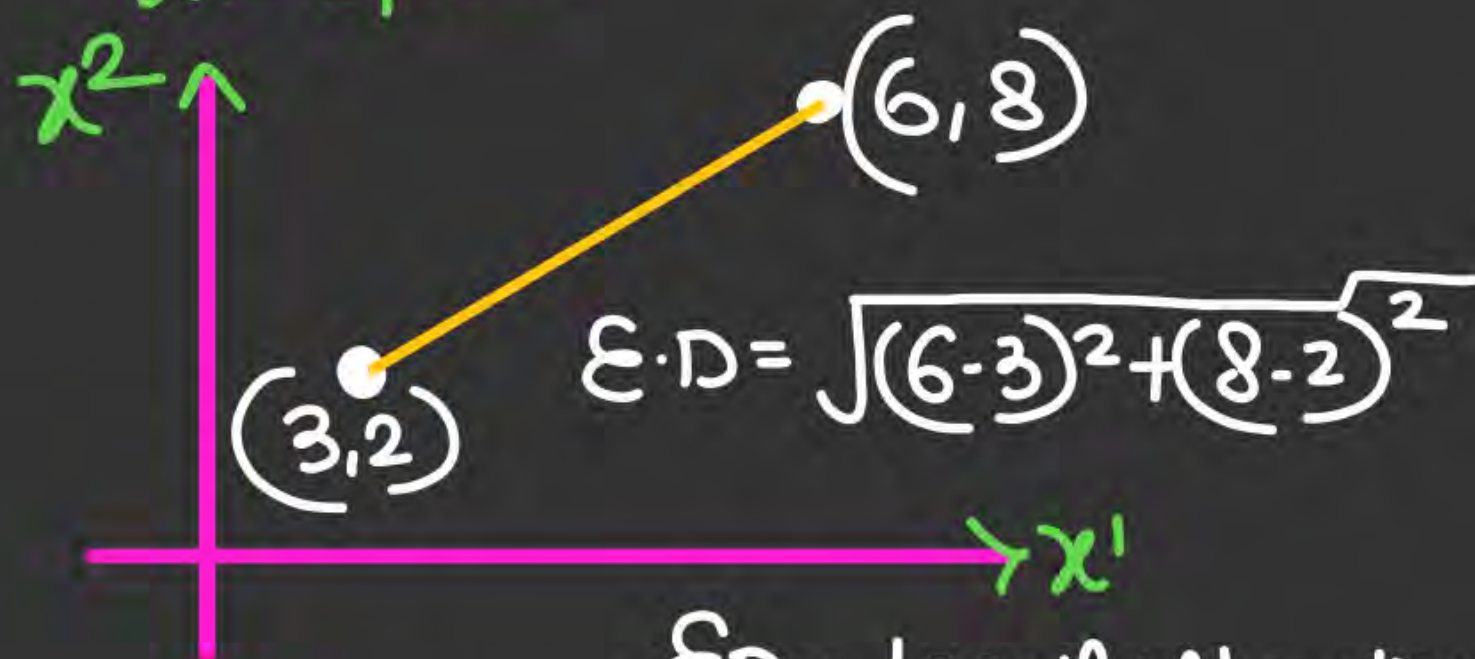
$$d_{12} = \sqrt{\sum_{i=1}^D (x_1^i - x_2^i)^2}$$

Ch: 3

Nearest Neighbour algorithm

NN: algorithm...

2D space



ED = length of line directly
Joining the points.

3D....

$$x_1 = (x_1^1, x_1^2, x_1^3)$$

$$x_2 = (x_2^1, x_2^2, x_2^3)$$

Euclidean distance...

$$d_{12} = \sqrt{(x_1^1 - x_2^1)^2 + (x_1^2 - x_2^2)^2 + (x_1^3 - x_2^3)^2}$$

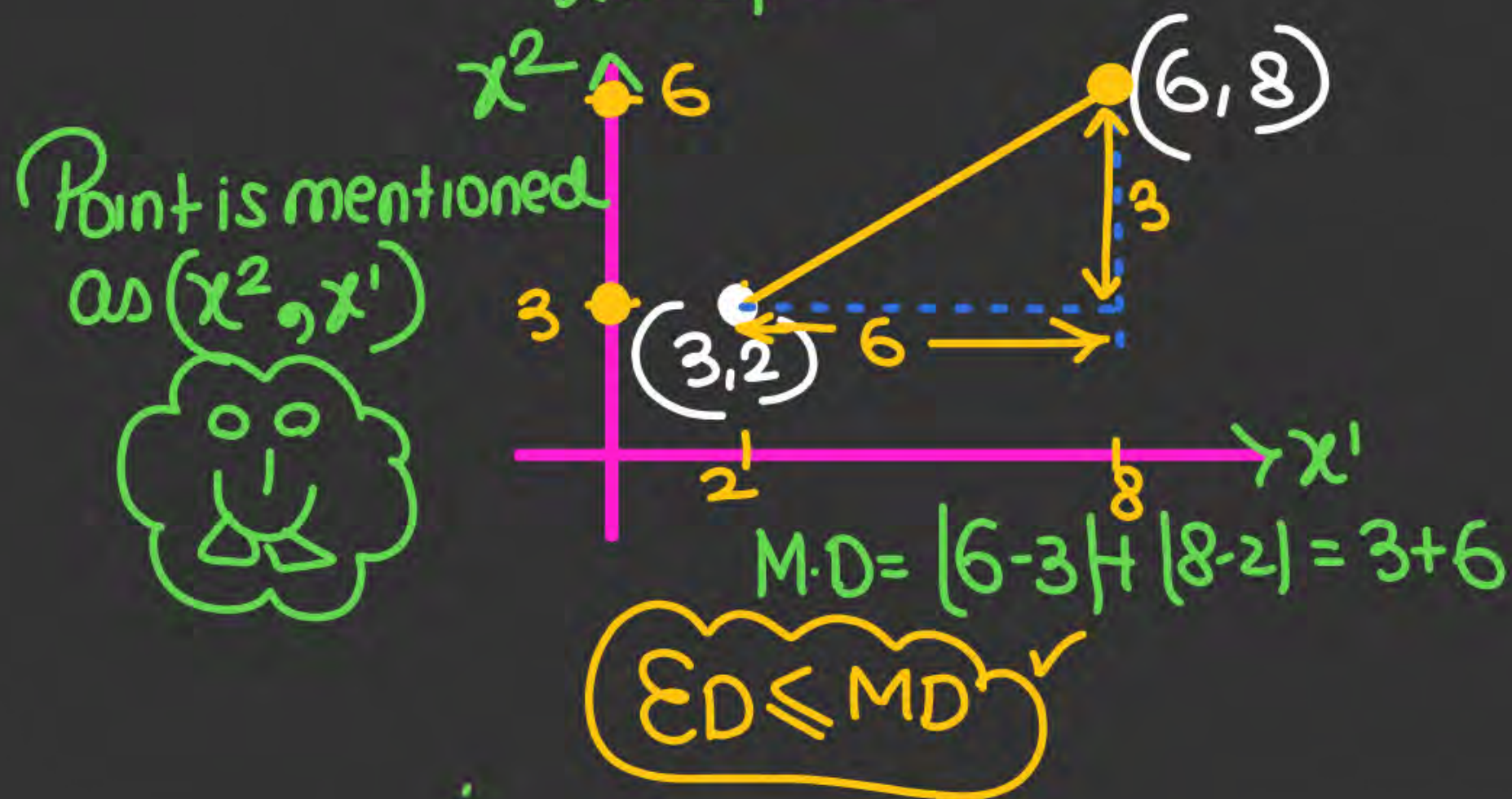
$$d_{12} = \sqrt{\sum_{i=1}^D (x_1^i - x_2^i)^2}$$

Ch: 3

Nearest Neighbour algorithm

NN: algorithm...

2D space



3D....

$$x_1 = (x_1^1, x_1^2, x_1^3)$$

$$x_2 = (x_2^1, x_2^2, x_2^3)$$

Manhattan distance...

$$d_{12} = |x_2^1 - x_1^1| + |x_2^2 - x_1^2| + |x_2^3 - x_1^3|$$

$$d_{12} = \sum_{i=1}^D |(x_1^i - x_2^i)|$$

In NN algorithm: distance is
Similarity measure....

algo Can be used for
Classification & regression
both.....
No Condition only.

NN algorithm: The lazy
algorithm..

Main data

1	x_1^1	x_1^2	y_1
2	x_2^1	x_2^2	y_2
3	x_3^1	x_3^2	y_3
4	x_4^1	x_4^2	y_4
...	...	...	...

algo
Not training
Process

We only keep
training data
ready.....

* In testing process we
Find distance of test point
From the training data
point....

* The point which is NN to
test point is Found...

Test point (x_t^1, x_t^2)
Point Jiska y Predict
Karne hai....

$y_{\text{predicted of Test point}} = y_{\text{value of NN}}$



Nearest Neighbour Method



K-NN method of Supervised Learning

The general flow of the Supervised Learning Algorithms

dekh lo... bas.

Instance
Based
Learning



Nearest Neighbour Method



Nearest Neighbour method of Supervised Learning

But in Nearest Neighbour the process is as follows...

- Linear Reg
 - Ridge Reg
 - Linear cl...
 - Logistic Reg...
- We put a Condition that... data shd come from a linear eq...
- We have a Condition that data shd be linearly Classifiable...

This is non parametric in nature

• In NN algo we do not have any Condition on what shd be the underlying Pattern of the data...



Nearest Neighbour Method



Nearest Neighbour method of Supervised Learning

But in Nearest Neighbour the process is as follows...

- * So data can come from any pattern...
- * we are not here to find any Equation relating y and x ✓

This is non
parametric in
nature

In NN algo we do not have any Condition on what should be the underlying pattern of the data...

Parametric algo:-

In Linear regression

$$\underset{\text{Actual}}{y} = \underset{\text{Predicted}}{wx} + \epsilon$$

So we assumed that data
Follow a linear pattern
And linear regression inhi w's ko
find Karne Chahata hai

$$\hat{y} = wx$$

So in parametric algo
we assume that data is
Coming from a particular
Pattern and humen bas
us pattern ki equation ke
parameters ko find karne
hai

Question:—

Use ED

Find y for test point $(5, 3)$ $y=12$ ✓

	data	y
1	3, 4	12
2	2, 9	18
3	1, 6	28
4	4, 8	42
5	6, 12	96

$$\rightarrow \sqrt{(5-3)^2 + (4-3)^2} = \sqrt{4+1} = \sqrt{5}$$

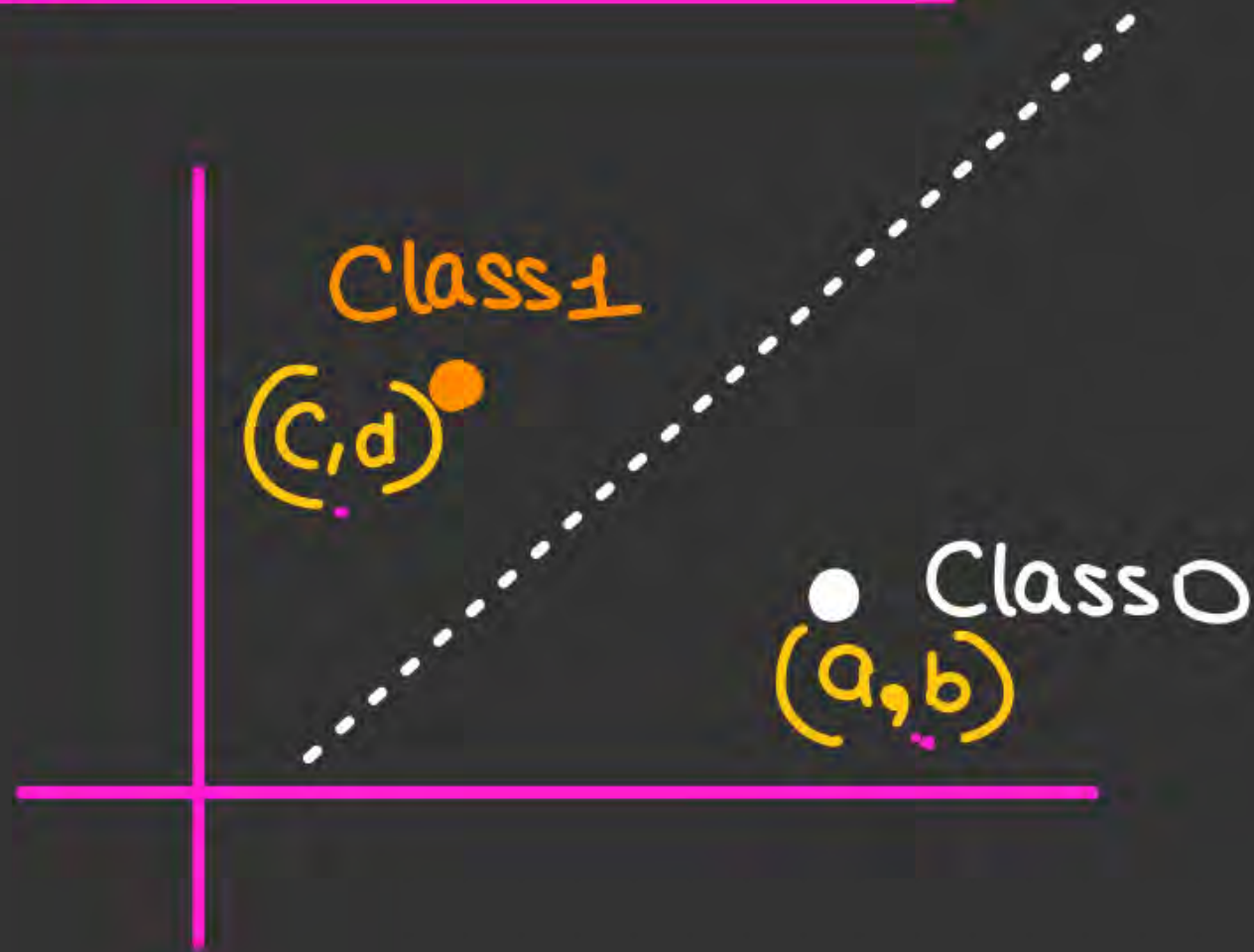
$$\rightarrow \sqrt{(5-2)^2 + (9-3)^2} = \sqrt{3^2 + 6^2} = \sqrt{45}$$

$$\rightarrow \sqrt{(5-1)^2 + (3-6)^2} = \sqrt{4^2 + 3^2} = \sqrt{25}$$

$$\rightarrow \sqrt{(5-4)^2 + (8-3)^2} = \sqrt{1+5^2} = \sqrt{26}$$

$$\rightarrow \sqrt{(5-6)^2 + (3-12)^2} = \sqrt{1+9^2} = \sqrt{82}$$

Concept Classification problem:—



• Training data has only 2 points

• any new point (x^1, x^2)

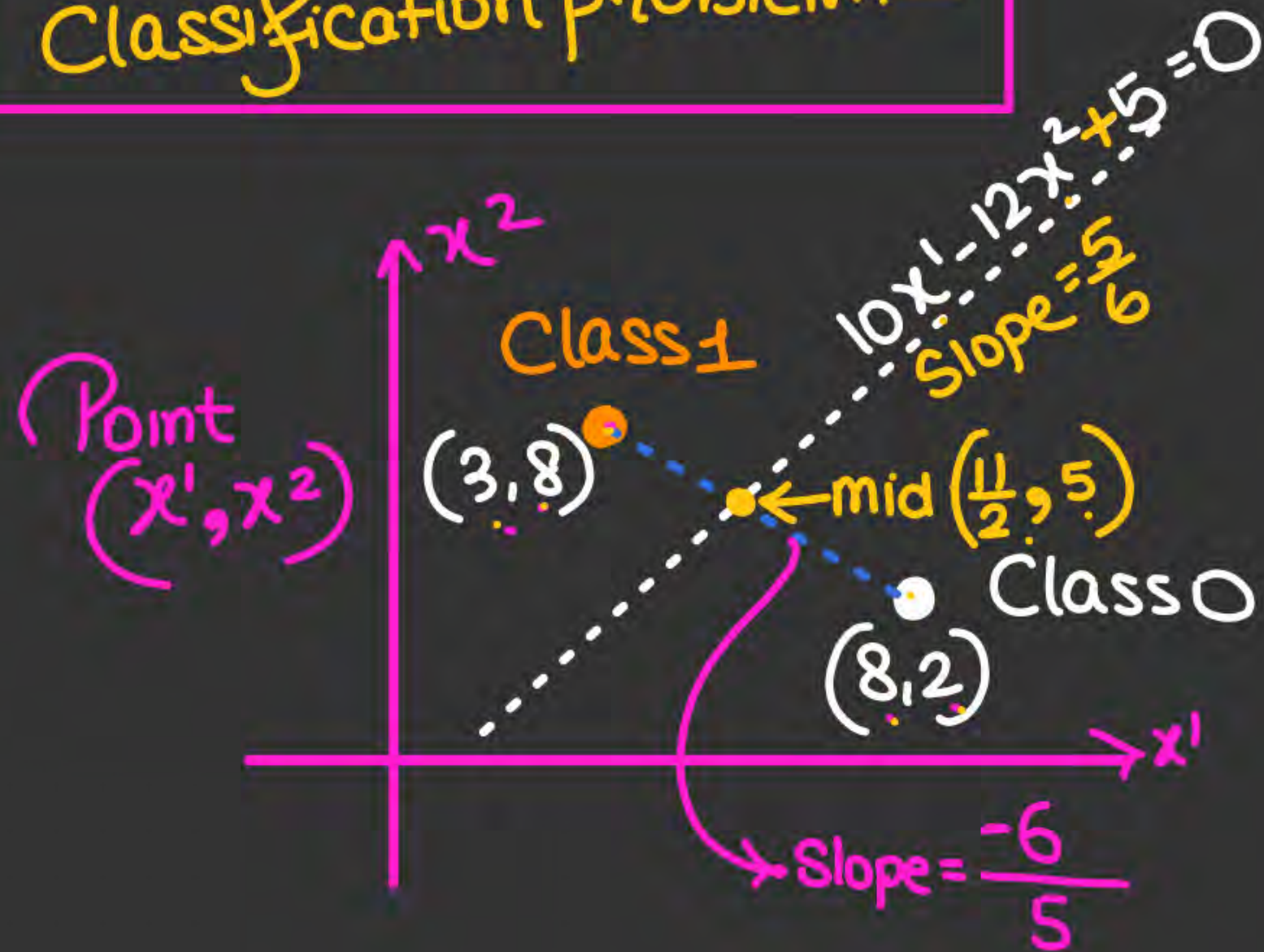
Rule distance of test point $F_{\text{nom}}(a, b)$ $\begin{matrix} > \\ < \end{matrix}$ distance of test point $F_{\text{nom}}(c, d)$

$$\sqrt{(x^1 - a)^2 + (x^2 - b)^2} \stackrel{\text{boundary}}{=} \sqrt{(x^1 - c)^2 + (x^2 - d)^2}$$

$$\begin{aligned} \cancel{(x^1)^2} + a^2 - 2ax^1 + \cancel{(x^2)^2} + b^2 - 2bx^2 &= \cancel{(x^1)^2} + c^2 - 2cx^1 + \cancel{(x^2)^2} + d^2 - 2dx^2 \\ \left[(2c - 2a)x^1 + (2d - 2b)x^2 \right] &= c^2 + d^2 - a^2 - b^2 \end{aligned}$$

Straight line eq. that act as a classifier...

Concept Classification problem:-



• Training data has only 2 points

• any new point (x^1, x^2)

Rule distance of test point $F_{\text{nom}}(a, b)$ $\begin{matrix} > \\ < \end{matrix}$ distance of test point $F_{\text{nom}}(c, d)$

$$(x^1 - 8)^2 + (x^2 - 2)^2 = (x^1 - 3)^2 + (x^2 - 8)^2$$

$$\cancel{64} - 16x^1 + 4 - 4x^2 = 9 - 6x^1 + \cancel{64} - 16x^2$$

$$10x^1 - 12x^2 = -5$$

$$[10x^1 - 12x^2 + 5 = 0]$$

$$x^2 = \frac{10}{12}x^1 + \frac{5}{12}$$

Two points only: NN algo

→ classifier...



Analysis of NN algo



green

Pink

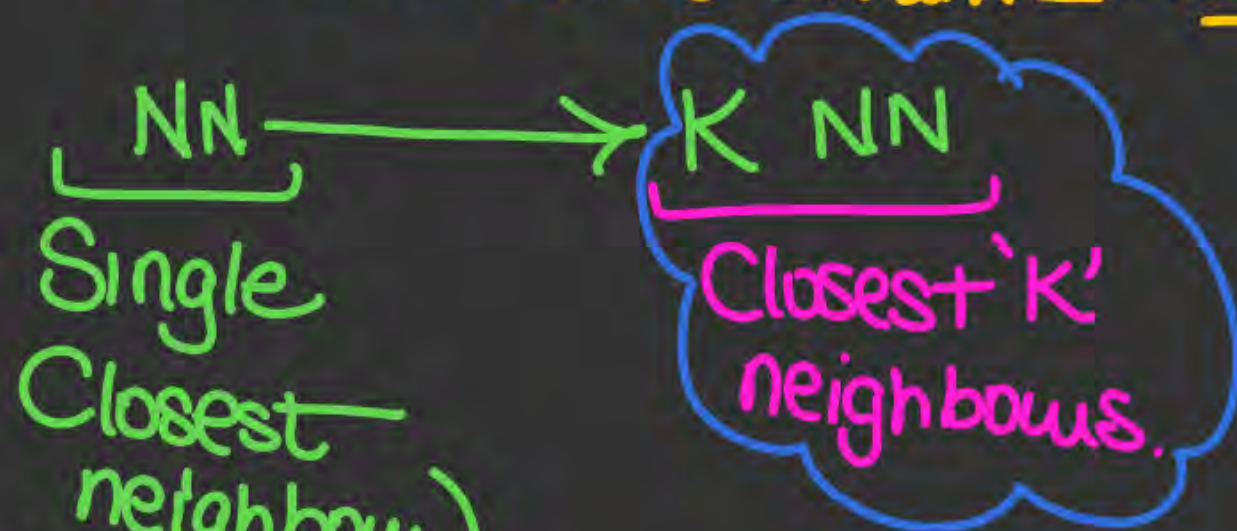
$$\cancel{\beta_0 + \beta_1 x + \beta_2 x^2}$$

- overfitting → unstable algo...
- every point has influence
- Complex model
- outlier point effect model.



Solution is $\Rightarrow$

Instead of giving importance to
a single point, single neighbour
to predict class of a test point
we consider more than 1 Neighbour



$\Rightarrow \hat{y} = y \text{ Value of closest Neighbour}$

K : Integer,
• Number of neighbours
we consider for
predicting y .

K NN

Closest 'K' neighbours.

How to find K??

Regression Case

y: Integers.

* Predicted y of test point

$\hat{y} = \text{avg } y\text{'s of all } K \text{ neighbours}$

$$\hat{y} = \frac{y_1 + y_2 + y_3 + \dots + y_K}{K}$$

Classification Case

y: Categorical nature...

Predicted y of test point

$\hat{y} = (\text{majority class of the } K \text{ nearest neighbours})$

example

3NN algo

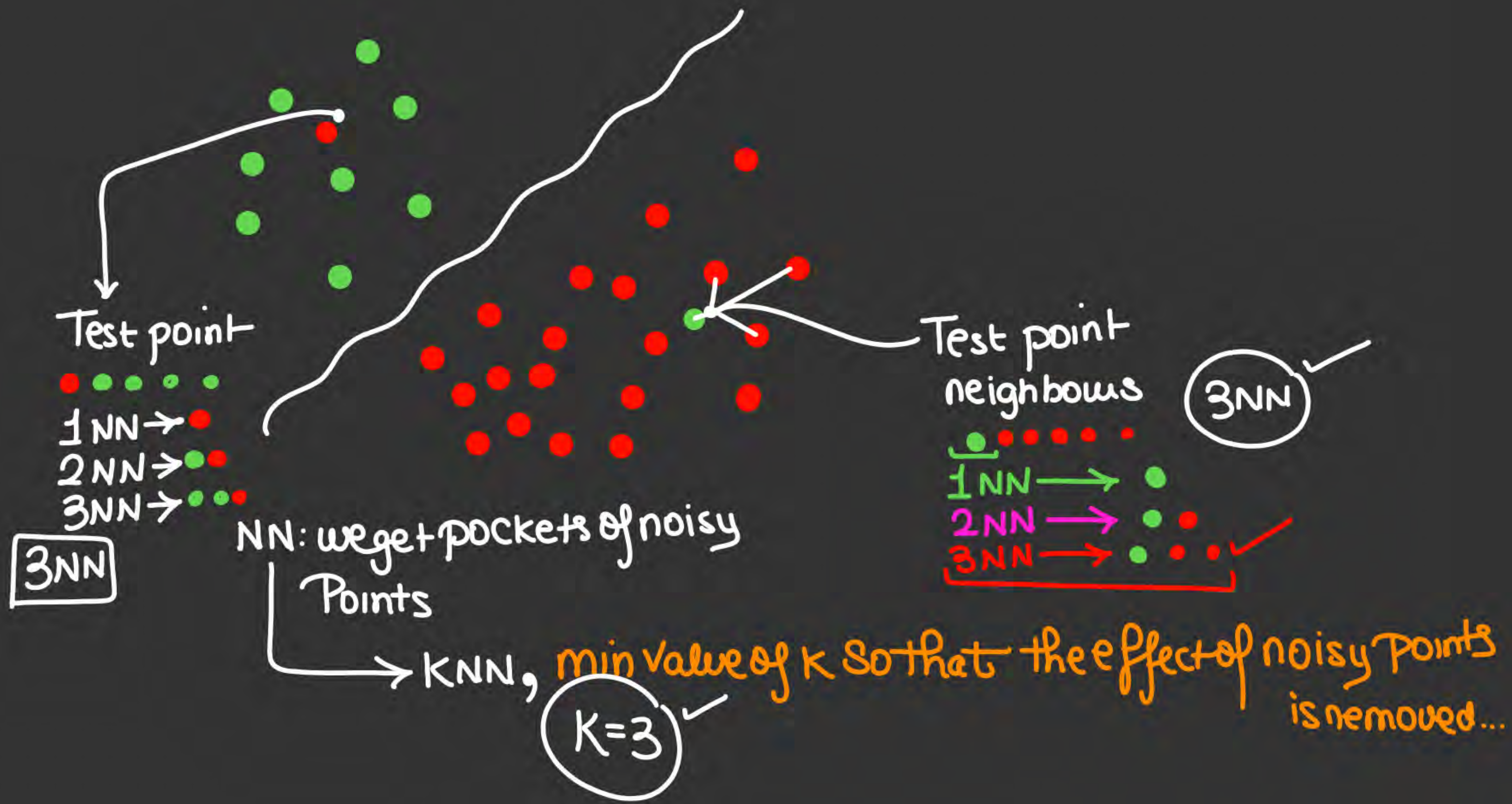
* 3 neighbours of a test point

$$y_1 = 10 \quad y_2 = 12 \quad y_3 = 11$$

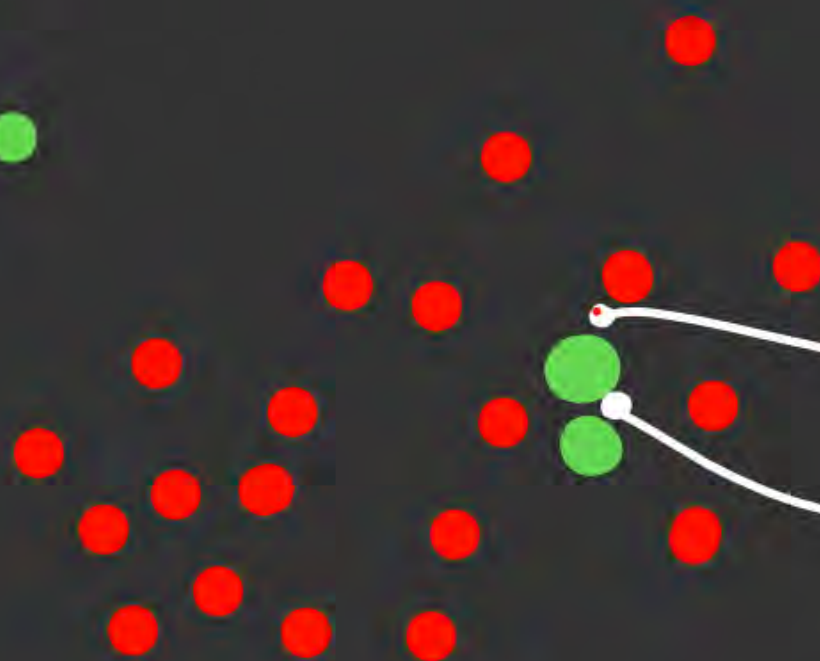
$$\hat{y} = \frac{10 + 12 + 11}{3} \Rightarrow \textcircled{11}$$

$$* y_1 = '1' \quad y_2 = '0', \quad y_3 = '0'$$

$$\hat{y} = \text{majority}(1, 0, 0) \\ = 0$$



Some Situation
K=3 Solve ✓



Solve problem @ worst
location

K=5

- 1NN •
- 2NN ••
- 3NN •••
- 4NN ••••
- 5NN •••••**

NN: we get pockets of noisy
Points

→ KNN, min value of K so that the effect of noisy points
is removed...

So

data 200 points

150 +
50 -

$\frac{1}{N}$
Overfit

$K=N$
Underfit

$\frac{1}{N}$
1NN algo
Overfit
algo

Under Fit
200
 K
for any test point
all the datapoints will
become neighbors....
 $\hat{y}$ = majority class
= (150 +, 50 -) $\Rightarrow$ (+)



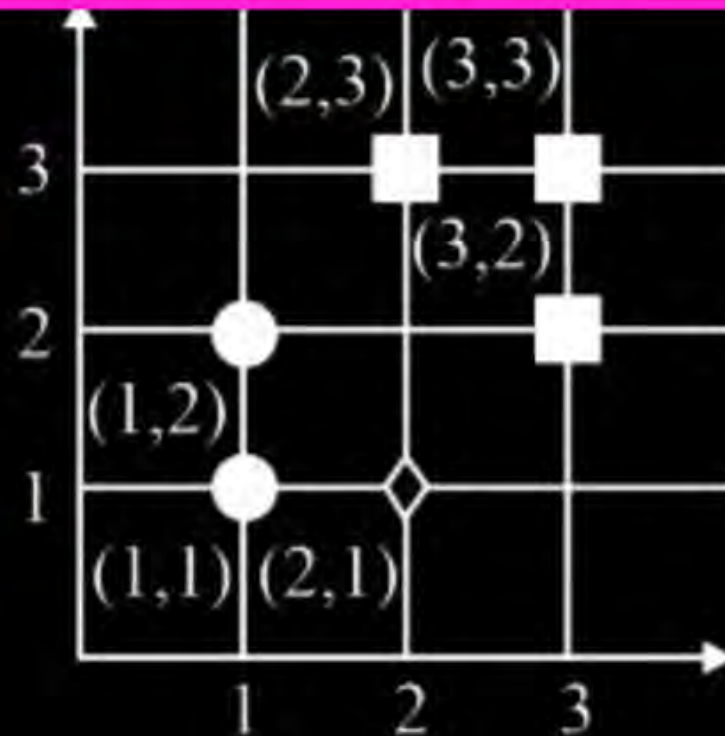
Nearest Neighbour Method



Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k -nearest neighbor algorithm for which the diamond ($\diamond$) shaped data point is assigned the label square is ____.

5NN ✓

- 1) 1, 2 $\sqrt{(2-1)^2 + (1-2)^2} = \sqrt{2}$ ✓
- 2) 1, 1 $\sqrt{(2-1)^2 + (1-1)^2} = 1$ ✓
- 3) 2, 3 $\sqrt{(2-2)^2 + (3-1)^2} = 2$
- 4) 3, 2 $\sqrt{(3-2)^2 + (2-1)^2} = \sqrt{2}$ ✓
- 5) 3, 3 $\sqrt{(3-2)^2 + (3-1)^2} = \sqrt{5}$



KNN

- 1NN $\Rightarrow$ 2nd $\Rightarrow$ ●
- 3NN $\Rightarrow$ 1, 2, 4 $\Rightarrow$ ● ● ▨
- 5NN $\Rightarrow$ 1, 2, 3, 4, 5 $\Rightarrow$ ● ● ▨ ▨ ▨



Nearest Neighbour Method



Nearest Neighbour method of Supervised Learning

What is K-NN Method of Supervised Learning

KNN
* Solve problem of overfitting

* make model more stable....✓

* avg $\Rightarrow \frac{a+b+c}{3}$

weighted avg } we assign weight to each
number, more imp $\rightarrow$ big weight
less imp $\rightarrow$ Small weight

KNN

- * Solve problem of overfitting
- * make model more stable....✓

* $\text{avg} \Rightarrow \frac{a+b+c}{3}$

weighted avg

we assign weight to each

number, more imp $\rightarrow$ big weight

less imp $\rightarrow$ Small weight

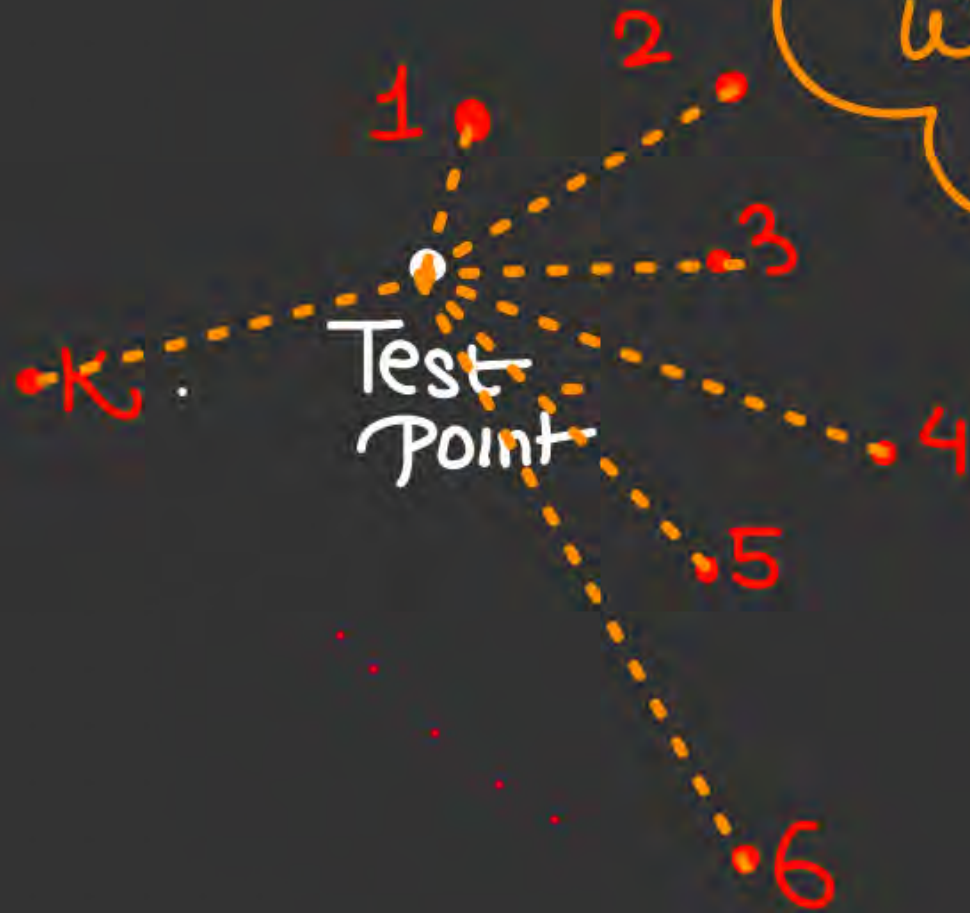
$$\frac{\omega_a \times a + \omega_b \times b + \omega_c \times c}{\omega_a + \omega_b + \omega_c}$$

Mention hoga
Kya lena hai...

$$\omega_i = \frac{1}{D_i}$$
$$\omega_i = \frac{1}{D_i^2}$$

we have K neighbours to test point

$$\hat{y} = \frac{y_1 + y_2 + \dots + y_K}{K}$$



- weighted KNN $\Rightarrow$ neighbour which is closer given more importance....
neighbour are given less importance...

- So weight of each neighbour $\omega_i = \frac{1}{D_i^2}$
 D_i = distance b/w test point and i^{th} neighbour

$$\hat{y} = \frac{\omega_1 y_1 + \omega_2 y_2 + \omega_3 y_3 + \dots}{\omega_1 + \omega_2 + \omega_3 + \dots}$$

USE 3NN, $w_i = \frac{1}{D_i}$

Find y for test point $(5, 3)$

Question:—

	data	y
1	3, 4	12
2	2, 9	18
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4	4, 8	42
5	6, 12	96

$$\rightarrow \sqrt{(5-3)^2 + (4-3)^2} = \sqrt{4+1} = \sqrt{5}$$

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$$\rightarrow \sqrt{(5-4)^2 + (8-3)^2} = \sqrt{1+5^2} = \sqrt{26}$$

$$\rightarrow \sqrt{(5-6)^2 + (12-3)^2} = \sqrt{1+9^2} = \sqrt{82}$$

$$\hat{y} = \frac{1}{D_1} y_1 + \frac{1}{D_3} y_3 + \frac{1}{D_4} y_4$$
$$= \frac{\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{26}} + \frac{1}{\sqrt{25}}}{\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{26}} + \frac{1}{\sqrt{25}}} = 22.77$$



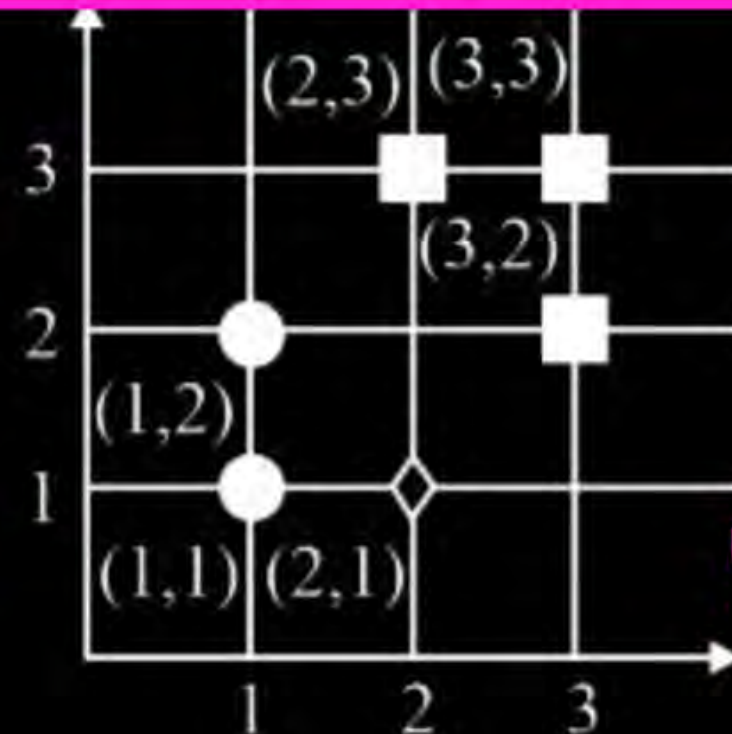
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5NN ✓

- 1) 1, 2 $\sqrt{(2-1)^2 + (1-2)^2} = \sqrt{2}$ ✓
- 2) 1, 1 $\sqrt{(2-1)^2 + (1-1)^2} = 1$ ✓
- 3) 2, 3 $\sqrt{(2-2)^2 + (3-1)^2} = 2$
- 4) 3, 2 $\sqrt{(3-2)^2 + (2-1)^2} = \sqrt{2}$ ✓
- 5) 3, 3 $\sqrt{(3-2)^2 + (3-1)^2} = \sqrt{5}$



Weighted 5NN $\Rightarrow$

$$\hat{y} = \frac{\sum y_i^o w_i^o}{\sum w_i}$$

$$= \frac{\frac{1}{\sqrt{2}} \bullet + \frac{1}{1} \bullet + \frac{1}{2} \square + \frac{1}{\sqrt{2}} \square + \frac{1}{\sqrt{5}} \square}{\frac{1}{\sqrt{2}} + 1 + \frac{1}{2} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{5}}}$$

$$\hat{y} = \frac{1.707 \bullet + 1.654 \square}{3.36}$$

So in Case of classification

$\hat{y}$ = Class with bigger Coefficient
in the equation...



Nearest Neighbour Method

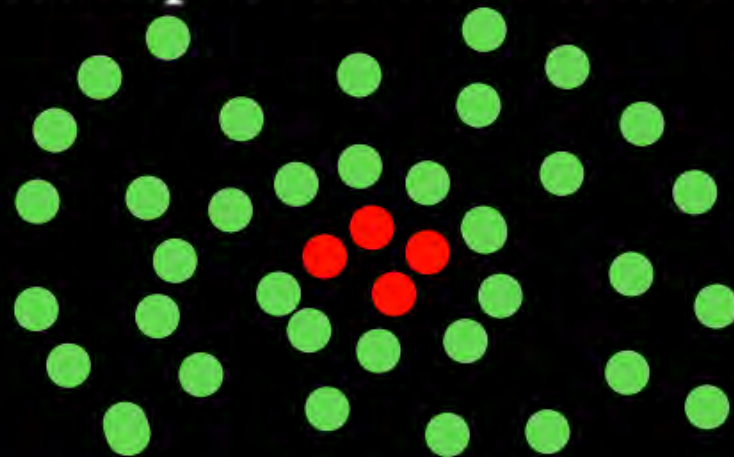


Nearest Neighbour method of Supervised Learning

Some Questions

False.

Is it possible for a 2-class 1-Nearest Neighbor (1-NN) classifier to always classify all new examples as positive even though there are negative examples in the training data? If yes, provide an example to illustrate this scenario. If no, briefly explain why it is not possible.



• Can I assume that always all new examples $\rightarrow$ $\bullet$
1NN



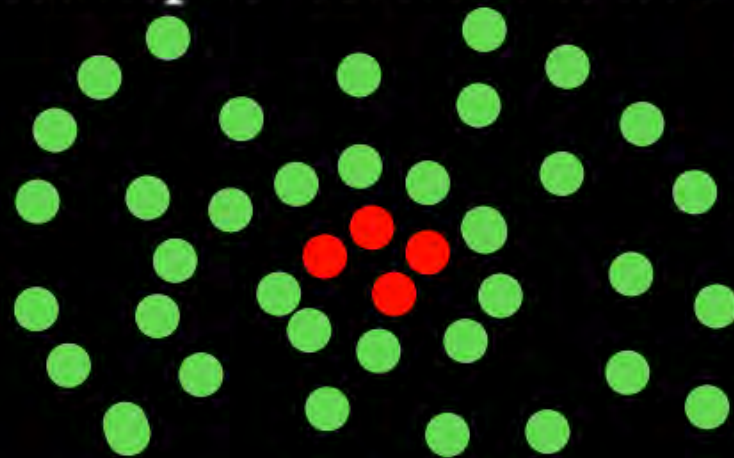
Nearest Neighbour Method



Nearest Neighbour method of Supervised Learning

Some Questions

Is it possible for a 2-class K -Nearest Neighbor (K -NN) classifier to always classify all new examples as positive even though there are negative examples in the training data? If yes, provide an example to illustrate this scenario. If no, briefly explain why it is not possible.



• Can I assume that always all new examples $\rightarrow$ 
 $K=9 \rightarrow$ always all test point green..

KNN

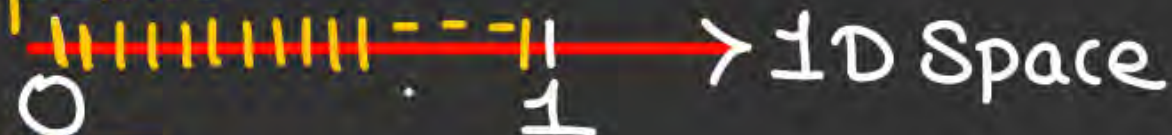
- * Very good algo
- * Easy & Simple
- * Nonparametric
- * Both regression + Classification...

Problems

① Curse of dimensionality and scale of dimensions $\Rightarrow$

1D data, Range of dimension $0 \rightarrow 1$

100 points

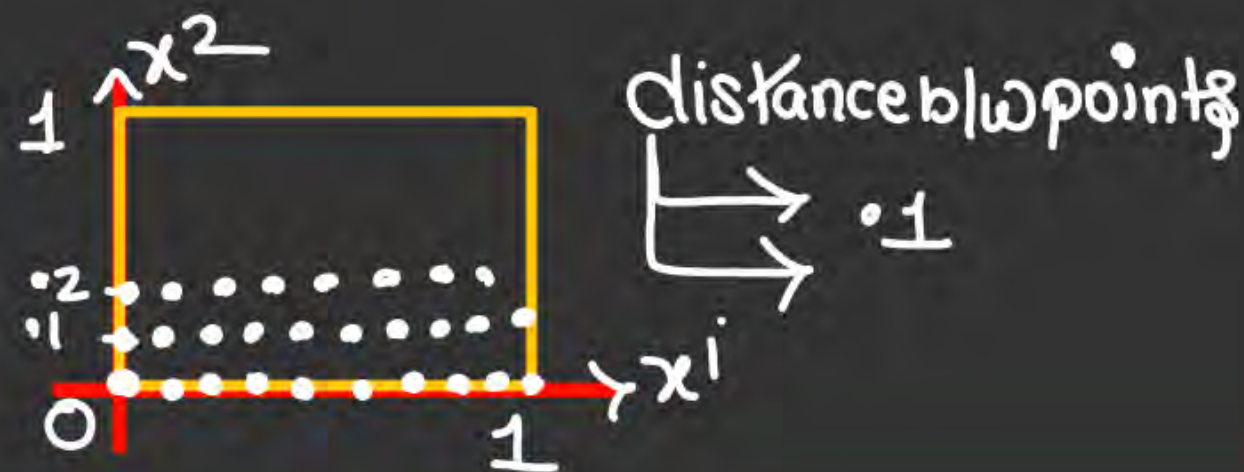
 $\rightarrow$ 1D Space

0

1

0.01

2D data
100 points



if No of data points remain same

↓
* we inc dimensions of data

* distance b/w points inc →

- KNN: depend on good neighbours
Similar ~



if No of dimension ↑ ⇒ distance b/w Points ↑ ⇒ So the space become sparse ⇒ Now neighbours will be Far apart, effecting Similarity...

Problem is if dimensions of data inc.

→ accuracy of KNN dec

* Similar effect of scale of the dimension *



Solution

1) If data has Large no of dimension
or range of dimension is high $\Rightarrow$
we can take more data

2) Scale the dimension of data...

3) reduce
dimension of
data...

Scaling dimension
all values convert
to range 0 $\rightarrow$ 1

data x^1 $\xrightarrow{\text{data modify}}$ $\frac{x^1 - x_{\min}^1}{x_{\max}^1 - x_{\min}^1} = \frac{x' - 5}{100 - 5} = \frac{x' - 5}{95}$

5	$\rightarrow$	$5 - 5 / 95 = 0$
15	$\rightarrow$	$15 - 5 / 95 = 10 / 95$
25	$\rightarrow$	$25 - 5 / 95 = 20 / 95$
85	$\rightarrow$	$85 - 5 / 95 = 80 / 95$
90	$\rightarrow$	$90 - 5 / 95 = 85 / 95$
100	$\rightarrow$	$100 - 5 / 95 = 1$

• Problem :- Computationally extensive
→ For every test point we have to find distance of test point with all training data points then Find KNN, then find $\hat{y}$,

* Linear Reg,
Linear class
• we only store β 's
after analysis, delete
training data...
• In KNN training data
Can never be deleted..

[MCQ]



#Q. Which of the following is true about the K-Nearest Neighbors (KNN) algorithm?

- ☒ **A** It is a parametric algorithm.
 - ☒ **B** It requires a large amount of memory to store the entire training dataset.
 - ☒ **C** It is sensitive to the scale of the data.
 - ☒ **D** Both b and c.
- Handwritten notes:*
A large 'X' is drawn over option A.
A checkmark is drawn over option B.
A checkmark is drawn over option C.
A checkmark is drawn over option D.
The text "Range dimension of" is written in yellow above option C, with an arrow pointing to the word "scale".
The text "b, c" is circled in yellow, with an arrow pointing to the checkmark on option D.

[MCQ]



#Q. In KNN, what happens when the value of K is set to 1?

NN : overfit, unstable,
Complex
Outlier

- A** The model becomes ~~more~~ robust to noise.
- B** The model is likely to overfit.
- C** The decision boundary becomes smoother.
- D** The model requires less memory.

[MCQ]



#Q. What is the effect of increasing the value of K in KNN?

$K=1 \rightarrow N$ model becomes less complex, more smooth

A

The model becomes more complex.

bad

B

The model becomes less sensitive to noise.

C

The decision boundary becomes more jagged.

D

The model tends to underfit.

[MCQ]



#Q. Which of the following is a disadvantage of KNN?

- A** It is computationally expensive for large datasets.
- B** It requires feature scaling.
- C** It is not suitable for high-dimensional data.
- D** All of the above.

#Q. Given the following training data points with their corresponding classes:

Feature 1	Feature 2	Class
1	2	A
2	3	A
3	4	B
4	5	B

Using KNN with $K = 3$ and Euclidean distance, classify the new data point $(2.5, 3.5)$. H.W. (A)

[MCQ]



#Q. Which of the following statements are true for k-Nearest Neighbors (k-NN) classifiers? (Circle all answers that are correct.)

- ☐ **A** The classification accuracy is better with larger values of k
- ☐ **B** The decision boundary is smoother with smaller values of k
- ☒ **C** k-NN is a type of instance-based learning
- ☐ **D** k-NN does not require an explicit training step
- ☐ **E** The decision boundary is linear

[MCQ]



#Q. K-Nearest Neighbor is a _____, _____ algorithm

A

Non-parametric, eager

B

Parametric, eager

C

Non-parametric, lazy

D

Parametric, lazy



[MCQ]



#Q. Which of the following distance metric can not be used in k-NN?

A Manhattan

B Minkowski

C Tanimoto

D Jaccard

E Mahalanobis

F All can be used



Thank you.

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